

Nicola Muca Cirone

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EDUCATION	PhD Mathematics - CDT Mathematics of Random Systems, 2022-2025 Imperial College London - University of Oxford • Deep Learning, Mathematical Finance, Rough Paths Analysis. • Supervised by <i>Dr. Cristopher Salvi</i>. • Researching scaling limits of Neural Networks • Working in theoretical foundations for State Space Models. • Developing new Kernel Methods for Time Series analysis. MSc Mathematics, University of Pisa - ETH Zürich 2021-2022 • Honour's degree (110/110 cum Laude) GPA 30 cum Laude / 30 • Stochastic Calculus, Stochastic PDEs, Mathematical Finance, ML in Finance. • Thesis: <i>Kernel Methods on Path Spaces</i> supervised by <i>Prof. Josef Teichmann</i>. BSc Mathematics, University of Pisa 2018-2021 • Honour's degree (110/110 cum Laude) GPA 29.88/30 • Thesis on advanced topics in Geometrical Model Theory.
EXPERIENCE	Intern - Quant Researcher, Quadrature Capital London July - Sept 2025 Deep learning for financial markets. Intern - ML Researcher, Cartesia AI San Francisco Feb - May 2025 Research on new ways to distill LLMs to different architectures, depths and sizes. Research on development of ViT models capable of multi-modal reasoning. Industry Project - Neural Signature Kernels, Imperial College London June 2024 Co-organized 2024 CDT Industry Project and supervised PhD students in applying Neural Signature Kernels to time series analysis. Teaching Assistant - Stochastic Calculus for Finance, ICL Business School Oct - Dec 2023 <i>MSc Risk Management & Financial Engineering</i>. Taught exercise classes. Industry Project - Brain signal analysis with Signatures, Imperial College London June 2023 Challenge: analyze MEG data from several subjects, detect the one who had a seizure. Our team used Signature techniques, the success and clarity of our results won the competition. Teaching Assistant - Numerical Methods for Finance, Imperial College London Feb 2023 <i>MSc Mathematics and Finance</i>. Taught exercise classes. Teaching Assistant - Data Analysis, Università di Pisa 2021-2022 <i>MSc Mathematics</i>. Provided tutoring support.
REVIEWED PUBLICATIONS	[2] Nicola Muca Cirone, Antonio Orvieto, Benjamin Walker, Cristopher Salvi and Terry Lyons. <i>Theoretical Foundations of Deep Selective State-Space Models.</i> <i>Accepted to NeurIPS 2024</i> url: https://arxiv.org/abs/2402.19047 [1] Nicola Muca Cirone, Maud Lemerrier and Cristopher Salvi. <i>Neural signature kernels as infinite-width-depth-limits of controlled ResNets.</i> <i>Proceedings of the 40th International Conference on Machine Learning</i>, 25358–25425, 2023. url: https://proceedings.mlr.press/v202/muca-cirone23a.html
PREPRINTS	

- [7] **Nicola Muca Cirone**, Owen Futter and Blanka Horvath.
Kernel Learning for Mean-Variance Trading Strategies
url: <https://arxiv.org/abs/2507.10701>
- [6] Benjamin Walker, Lingyi Yang, **Nicola Muca Cirone**, Cristopher Salvi and Terry Lyons.
Structured Linear CDEs: Maximally Expressive and Parallel-in-Time Sequence Models.
url: <https://arxiv.org/abs/2505.17761>
- [5] Sajad Movahedi, Felix Sarnthein, **Nicola Muca Cirone** and Antonio Orvieto.
Fixed-Point RNNs: From Diagonal to Dense in a Few Iterations
url: <https://arxiv.org/abs/2503.10799>
- [4] **Nicola Muca Cirone** and Cristopher Salvi.
ParallelFlow: Parallelizing Linear Transformers via Flow Discretization.
url: <https://arxiv.org/abs/2504.00492>
- [3] Mie Glückstad, **Nicola Muca Cirone** and Josef Teichmann.
Signature Reconstruction from Randomized Signatures.
Submitted to Journal
url: <https://arxiv.org/abs/2502.03163>
- [2] **Nicola Muca Cirone** and Cristopher Salvi.
Rough kernel hedging.
url: <https://arxiv.org/abs/2501.09683>
- [1] **Nicola Muca Cirone**, Jad Hamdan and Cristopher Salvi.
Graph Expansions of Deep Neural Networks and their Universal Scaling Limits.
Submitted to Journal
url: <https://arxiv.org/abs/2407.08459>

TALKS	Oxford-ETH Workshop in Mathematical Finance , Oxford University Title: <i>Signature Reconstruction from Randomized Signatures.</i>	June 2025
	Workshop on Recent Developments in Theoretical Machine Learning , Imperial-X Title: <i>Graph Expansions of Deep Neural Networks.</i>	Jan 2025
	Stochastic Analysis Seminar , ShanghaiTech University Title: <i>Theoretical Foundations of Deep Selective State-Space Models.</i>	Nov 2024
	DataSig - Rough Paths Interest Group , Alan Turing Institute Title: <i>Signature Reconstruction from Randomized Signatures.</i>	May 2024
	Computer Vision Group Seminar , University of Michigan Title: <i>Theoretical Foundations of Deep Selective State-Space Models.</i>	April 2024
	Talks in Financial and Insurance Mathematics , ETH Zürich Title: <i>Neural Signature Kernels (and Trees).</i>	Dec 2023
	7th London-Paris Bachelier Workshop , Imperial College London Title: <i>Rough Kernel Hedging.</i>	Sept 2023
AWARDS	NeurIPS 2024 Scholar Award , Neural Information Processing Systems Foundation	2024
	INDAM Scholarship , Istituto Nazionale di Alta Matematica Two year scholarship from the Italian Institute of Higher Mathematics. Placed 2nd at National level.	2022
COMPETENCES	Languages Italian (<i>native</i>), English (C1), French (<i>basic</i>)	
	Technical Skills Python, Jax, Pytorch, Triton, CUDA, R, Matlab, Java	